Low-resource Authorship Style Transfer via Dynamic Style Inference and Parameter Modulation

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Abstract

Low-resource authorship style transfer (LAST) aims to rewrite text in the style of an arbitrary target author using only a few reference examples while preserving the original meaning. Existing methods often struggle to achieve both high style fidelity and semantic preservation because they compress diverse references into a single static author embedding, which averages out context-dependent stylistic variation, and rely on hidden representations for style control, which entangle style with content. We propose HyperStyler, a novel architecture that decouples LAST into style inference and style realization. Stylo-navigator predicts context-dependent style coordinates by jointly modeling the source context and target-author references, while Stylo-hypernet realizes them via dynamic parameter modulation instead of hidden-state injection. By shifting authorship control to the parameter space, HyperStyler enables more faithful and controllable style transfer with reduced semantic interference. Experiments on Reddit, Blog, and News show that HyperStyler consistently outperforms prior methods, generalizes robustly across domains, and requires only a 2.4% increase in parameters.

1 Introduction

Even when writing the same content, individuals exhibit distinctive lexical choices, syntactic structures, and modes of expression (Stamatatos, 2009; Wang et al., 2023). Low-resource authorship style transfer (LAST) aims to rewrite a source text in the style of an arbitrary target author, preserving the original semantics given only a few reference examples (Patel et al., 2022). Unlike traditional style transfer, which is often restricted to authors with massive corpora, LAST extends the scope to everyday writers with a small number of sentences. This shift offers significant practical value, enabling users to personalize generic AI-generated content or efficiently transform drafts into the nuanced voice of any desired author.

Despite its potential, high-fidelity style transfer remains a significant challenge. Initial attempts leverage in-context learning (Patel et al., 2022) or inference-time control methods (Khan et al., 2023; Horvitz et al., 2024a), but often yield weak stylistic transfer. More recent approaches (Liu et al., 2024; Horvitz et al., 2024b) introduce unsupervised style alignment frameworks by constructing pseudo-paired datasets. However, they still struggle to achieve strong style fidelity and semantic preservation simultaneously.

According to the stylometry literature, an author's style is not a static template but a multifaceted phenomenon that shifts with topic and register (Sapkota et al., 2014; Hoover, 2017; Gracheva and Egbert, 2022; Grieve, 2023). Each reference captures how the target author writes in a particular context. In high-resource settings, abundant data allows models to implicitly learn an individual's stylistic variation across contexts. In the low-resource setting, however, such learning is no longer possible, and the model must rely solely on these references to infer and realize the author's context-dependent stylistic variation. Yet existing methods condition generation on a single static author embedding by compressing references (many-to-one mapping). Forcing diverse stylistic cues into a unified embedding induces mode averaging. This often causes the model to generate an indistinct *average* style, rather than the author's context-specific stylistic patterns, thereby limiting style fidelity.

Furthermore, controlling authorship through hidden representations often limits fine-grained controllability. When stylistic cues are injected directly into hidden states, they become entangled with semantic content, making author-specific patterns difficult to isolate and manipulate. This issue is particularly severe in LAST, where the model must generalize to arbitrary unseen authors rather than a fixed set of known authors. As a result, the model struggles both to perform precise stylistic transformations without distorting meaning and to capture each author's diverse stylistic variations.

To address these limitations, we propose HyperStyler, a novel architecture designed for high-fidelity LAST. HyperStyler explicitly decouples the task into inference and realization stages through two specialized modules. First, the Stylo-navigator predicts context-dependent style coordinates by considering both the input context and target-author references. This allows the model to select a distinct stylistic path that reflects the diverse variations of an author's style. Second, the Stylo-hypernet realizes these coordinates through dynamic parameter modulation rather than hidden-state injection. Shifting the control mechanism to the parameter space isolates style from semantics, thereby minimizing interference while improving steerability and integrity. Extensive experiments across diverse domains, including Reddit, Blog, and News, demonstrate that HyperStyler consistently outperforms existing baselines. Furthermore, our results confirm that our model achieves robust generalization and superior performance while requiring only a 2.4% increase in parameters, highlighting its efficiency and practical utility.

2 Related Work

2.1 Unsupervised Alignment for LAST

Due to the scarcity of parallel data, recent LAST methods commonly follow a two-stage unsupervised alignment framework (Krishna et al., 2020). The first stage trains a model to reconstruct the original text from style-neutralized paraphrases, conditioned on either a static author embedding (Horvitz et al., 2024b) or a set of reference samples (Liu et al., 2024). In the second stage, the model is further aligned using filtered pseudo-parallel data. HyperStyler also follows this framework; however, instead of aligning the model against a fixed author embedding, we align it toward predicted style coordinates that represent the most suitable stylistic variation for a given context. This approach allows HyperStyler to account for diverse stylistic nuances of the author.

2.2 Hypernetworks

Hypernetworks (Ha et al., 2016) generate parameters of a target model conditioned on an external signal, enabling more flexible modulation than

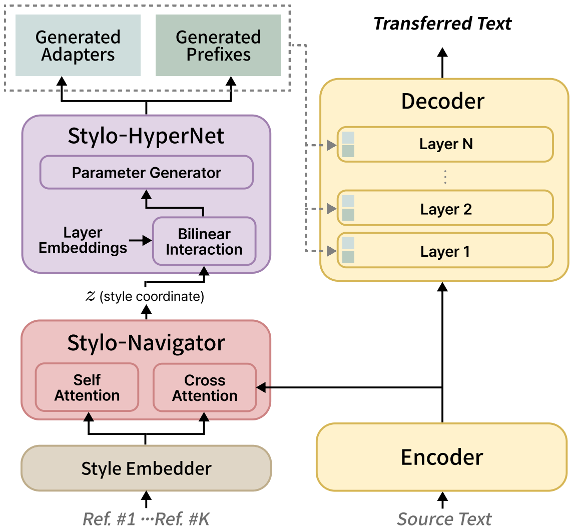

Figure 1: Overall architecture of HyperStyler

static adapters. In the natural language processing domain, they have been primarily used for task- or domain-conditioned adaptation (Ivison et al., 2023; Li et al., 2024). However, their use for parameter-space modulation in open-ended, few-shot style control, where the model must generalize to unseen conditioning signals and accurately capture context-specific stylistic facets, remains underexplored, especially for fine-grained linguistic patterns. In this paper, we condition hypernetwork-based parameter modulation on context-dependent stylistic coordinates derived from few-shot references. This enables scalable control in open-ended settings while reducing content-style entanglement and improving structural controllability under semantic preservation.

3 HyperStyler

3.1 Overview

Given a source text x and a few references $R = \{r_i\}_{i=0}^{K}$ written by a target author, our goal is to generate y that matches the target author's writing style while preserving the semantics of x. We explicitly decompose this process into two subtasks: (i) context-dependent style selection, which infers a style control signal from the references conditioned on the source context, and (ii) stylistic realization, which applies the inferred signal to rewrite x in the target style without altering its meaning.

As illustrated in Figure 1, HyperStyler decomposes LAST into two subtasks implemented by modules attached to an encoder–decoder paraphraser. We adopt this paraphraser as the underlying model because it facilitates decoupling content preservation in the encoder from stylistic realization in the decoder and aligns with established practices in the style transfer task (Krishna et al., 2020; Lee et al., 2021; Zhao et al., 2024; Horvitz et al., 2024b). First, the Stylo-navigator predicts a context-dependent style coordinate z from x and R. To ensure parameter efficiency, we reuse the backbone encoder representations $H_{\text{enc}} = \text{Enc}(x)$ as the contextual signal for x, avoiding an additional context encoder. Second, the Stylo-hypernet realizes the target style by mapping z to parameter modulations. These modulations are injected into the decoder via key/value prefixes and low-rank adapters in the feed-forward networks (FFNs).

3.2 Stylo-navigator

We define a stylistic coordinate z in the pretrained style embedding space using STYLE embedder (Wegmann et al., 2022), which is trained to capture content-independent stylistic representations. This reduces the influence of content semantics from the reference sentences on the control signal, encouraging z to primarily reflect stylistic characteristics. Each reference sentence r_i is mapped to a style embedding s_i, yielding a set of reference embeddings $S \in \mathbb{R}^{K \times d}$. This set is used as the input to the Stylo-navigator.

The Stylo-navigator predicts a stylistic coordinate z by conditioning the references on the source context via two parallel attention mechanisms[1]. First, we apply self-attention over S to capture the relational features that characterize the author's uniqueness, producing $\tilde{S} = \text{SelfAttn}(S)$. In parallel, cross-attention is applied with the backbone encoder hidden states H_{enc} as queries and the reference embeddings S as keys and values. This mechanism allows the model to select and weight reference styles differently depending on the source context. At the token level, each hidden state interacts with the entire references, enabling fine-grained, context-dependent stylistic selection. The resulting token-specific stylistic signals are aggregated via mean pooling to form a representative vector $q \in \mathbb{R}^d$, which summarizes the source-conditioned stylistic fitness:

$$q = \text{MeanPool}\big(\text{CrossAttn}(H_{\text{enc}}, S, S)\big). \quad (1)$$

To derive a final stylistic coordinate, we conduct a scaled-dot product with each reference style $\tilde{s_i} \in \mathbb{R}^d$ to compute α_i, which determines how much each reference embedding contributes to z:

$$\alpha_i = \frac{\exp(\bar{q} \cdot \bar{\tilde{s}}_i / \sqrt{d})}{\sum_{j=1}^{K} \exp(\bar{q} \cdot \bar{\tilde{s}}_j / \sqrt{d})}, \quad (2)$$

where d denotes the dimensionality of the style embedding space, and $\bar{q}$, $\bar{\tilde{s}}_i$ are layer-normalized for stability. The final stylistic coordinate is then obtained as the weighted sum of the reference embeddings:

$$z = \sum_{i=1}^{K} \alpha_i s_i. \quad (3)$$

Through this mechanism, the Stylo-navigator determines a coordinate z that represents a stylistic target reflecting both the source context and the author's stylistic uniqueness.

3.3 Stylo-hyperNet

Stylo-hyperNet dynamically modulates the decoder conditioned on a stylistic coordinate z. Prior analyses suggest that transformer layers contribute differently to generation, and stylistic realization can therefore be layer-dependent (Langedijk et al., 2024; Alshomary et al., 2025). Motivated by this observation and inspired by Ivison et al. (2023), we introduce learnable layer embeddings and condition them on z to construct layer-specific style signals. Concretely, we compute a style-dependent offset for each layer and add it to the corresponding layer embedding via a residual connection to preserve layer identity. The resulting layer-specific signals are then used to generate modulation parameters for subsequent decoder modulation.

Style-conditioned Layer Embeddings Let $E^{(t)} = [e_1; \ldots; e_{N_t}] \in \mathbb{R}^{N_t \times d_e}$ be a learnable embedding table associated with a modulation target t (e.g., cross-attention prefix keys/values or the up/down projections of an FFN adapter). Here, N_t is the number of embeddings for target t, and each row e_j corresponds to a distinct modulation-parameter instance indexed by a tuple (layer, type, position). For example, prefix embeddings are indexed by (layer, key/value, prefix position), while adapter embeddings are indexed only by projection type (layer, up/down).

Given a style coordinate z, we compute an embedding-wise modulation strength via a bilinear interaction:

$$s_j = (W_e \bar{e}_j)^\top (W_s \bar{z}), \quad (4)$$

[1]For reproducibility, we used a pre-norm attention structure.

where $W_e \in \mathbb{R}^{d \times d_e}$ and $W_s \in \mathbb{R}^{d \times d}$ are trainable projection matrices, and $\bar{e}_j$ and $\bar{z}$ denote layer-normalized vectors. The scalar s_j acts as a style-dependent coefficient for embedding e_j. We then form an offset by scaling a transformed style vector and projecting it with $W_v \in \mathbb{R}^{d \times d}$ to match the embedding space:

$$o_j = s_j \cdot (W_v \bar{z}). \tag{5}$$

Finally, we project the offset and apply layer normalization to obtain the style-conditioned update, which is added to the base embedding through a residual connection:

$$\Delta e_j = \mathrm{LN}(W_o\, o_j), \qquad \tilde{e}_j = e_j + \Delta e_j. \tag{6}$$

Generating Modulation Parameters. We map the style-conditioned embeddings $\tilde{E}^{(t)}$ to modulation parameters using two-layer MLPs. We use dedicated generators for different modulation targets, as each target operates on distinct parameter spaces with different dimensionalities and functional roles. Here, d_{model} denotes the hidden dimensionality of the underlying model.

Cross-attention prefixes modulate how the decoder references the source context during generation (Li and Liang, 2021). We generate key and value prefix vectors using two independent MLPs. The generated vectors are grouped into length-p prefixes per layer, $P_K^\ell, P_V^\ell \in \mathbb{R}^{p \times d_{\text{model}}}$, and concatenated to the original keys and values along the sequence dimension, $K'^\ell = [P_K^\ell; K^\ell]$ and $V'^\ell = [P_V^\ell; V^\ell]$.

Low-rank adapter in FFN modulates the FFN pathway. Since FFN layers have been shown to store substantial linguistic information (Geva et al., 2021, 2022), modulating this pathway can affect surface realization, including lexical choice and syntactic patterns. For each layer ℓ, we generate low-rank down- and up-projection weights using separate MLPs, $W_{down}^\ell \in \mathbb{R}^{d_{\text{model}} \times r}$ and $W_{up}^\ell \in \mathbb{R}^{r \times d_{\text{model}}}$. We add the resulting branch to the FFN output as a residual:

$$h_{\text{out}}^l = \mathrm{FFN}^l(h_{\text{in}}^l) + \sigma(h_{\text{in}}^l W_{down}^\ell) W_{up}^\ell, \tag{7}$$

where h_{in} and $\sigma(\cdot)$ denote the FFN input hidden states and the activation function, respectively. We choose σ to match the FFN activation in the underlying model. In our implementation, each MLP outputs a vector of length $r \cdot d_{\text{model}}$, which is reshaped into the corresponding low-rank matrix for each layer.

3.4 Training Procedure

Due to the lack of a parallel dataset for authorship style transfer, we adopt an unsupervised training setting, and the overall training procedure consists of three stages.

Stage 1: Training the Base Paraphraser. We train an encoder–decoder paraphraser as the underlying model of our framework. For each author, we collect an author-specific corpus consisting of a sentence set $X = \{x_i\}_{i=1}^K$. Using a pretrained paraphraser, we generate a synthetic paraphrase x_i' for each sentence x_i, thereby constructing synthetic parallel pairs $\{(x_i, x_i')\}_{i=1}^K$.

To mitigate stylistic bias inherited from the pretrained paraphraser and to improve both diversity and semantic preservation, we train the underlying paraphraser with a bidirectional reconstruction objective over $(x_i \leftrightarrow x_i')$ pairs (Sjöblom et al., 2020; Ma et al., 2021). The goal of this stage is not to acquire any specific style, but to establish a semantically reliable paraphrasing backbone.

Stage 2: Training the Stylo-navigator and Stylo-hypernet. In this stage, we freeze the underlying paraphraser and integrate it with the Stylo-navigator and Stylo-hypernet to be trained simultaneously through an unsupervised reconstruction task. For each author, the reference set is defined as their own corpus ($R = X$) and represented as a matrix $S \in \mathbb{R}^{K \times d}$.

Each training instance is designed to reconstruct the original sentence x_i from its paraphrased version x_i'. The Stylo-navigator is trained to identify the target stylistic reference within S that best matches x_i given the source context x_i'. We use the index i of the target sentence as a ground-truth label and minimize the negative log-likelihood of the predicted selection probabilities α:

$$\mathcal{L}_{\text{nav}} = -\log \alpha_i. \tag{8}$$

To prevent the navigator's initial prediction errors from propagating to the Stylo-hypernet, we employ a teacher-forcing strategy, providing the ground-truth style embedding s_i as input instead of the predicted coordinate z. This allows the Stylo-hypernet to focus on learning precise parameter modulation. The Stylo-hypernet is optimized to maximize the reconstruction likelihood of x_i:

$$\mathcal{L}_{\text{hypernet}} = -\sum \log p(x_i \mid x_i', s_i). \tag{9}$$

Stage 3: Training for Style Alignment. Finally, we optimize the model for realistic style transfer beyond simple reconstruction. In this stage, the Stylo-hypernet is conditioned on the stylistic coordinate z predicted by the Stylo-navigator. Since direct parallel data between source and target authors is unavailable, we construct a high-quality synthetic parallel dataset via self-distillation (Zhang et al., 2019b). Specifically, we generate style-transferred outputs using the model from Stage 2 and filter them following the *Rerank and Filtering* procedure from Horvitz et al. (2024b). Crucially, unlike prior work that evaluates style fidelity using a mean-pooled reference embedding, we filter generated sentences based on the z predicted by the Stylo-navigator. By performing supervised fine-tuning on this filtered dataset, we jointly optimize the Stylo-navigator and Stylo-hypernet, thereby reducing the discrepancy between style prediction and stylistic realization.

4 Experiments

4.1 Experimental Setup

Datasets. We conduct experiments on three datasets with distinct genres: **Reddit** (Million User Dataset; (Khan et al., 2021)), **Blog** (Schler et al., 2006), and **News**[2]. Following prior work of LAST (Patel et al., 2022; Horvitz et al., 2024a,b), we segment each author's corpus into sentences and randomly sample only 10 sentences per author. We filter out any samples exceeding 60 tokens and split the data by author into training, validation, and test sets with a 0.9/0.05/0.05 ratio. The detailed preprocessing procedures and statistics of the datasets are described in the Appendix.

We use three evaluation sets of **Reddit** from Patel et al. (2022): *Random*, *Single*, and *Diverse*. Each split comprises 15 source and 15 target authors with 16 samples each, totaling 225 transfer directions and 3,600 transformations. We apply the same configuration to **Blog** and **News** by randomly selecting authors from the hold-out authors in the test dataset.

Evaluation Metrics. To ensure fair and consistent comparison, we follow the evaluation protocol established in prior LAST studies (Patel et al., 2022; Horvitz et al., 2024b). We measure AWAY, TOWARDS, SIM, and JOINT. Specifically, AWAY and TOWARDS quantify stylistic shifts by measuring the degree to which the set of transferred texts moves away from the source author's style and toward the target author's style, respectively. These metrics are computed using a held-out UAR embedder (Rivera-Soto et al., 2021), which is trained via contrastive learning for large-scale authorship verification. To evaluate semantic preservation, we use the Mutual Implication Score (Babakov et al., 2022) as SIM. Finally, the JOINT score provides a summary of both style fidelity and semantic preservation, calculated as follows: $\text{JOINT} = G(G(\text{TOWARDS}, \text{AWAY}), \text{SIM})$, where $G(\cdot)$ denotes the geometric mean.

Baselines. We compare against baselines across three categories.

In-context learning methods include STYLL (Patel et al., 2022) with Qwen2.5, and models prompted with the instructions from Horvitz et al. (2024b), including Llama3.1, Qwen2.5, GPT4-turbo, and the reasoning enabled GPT5.4.

Inference-time control methods include ParaGuide (Horvitz et al., 2024a), a diffusion-based model, and StyleMC (Khan et al., 2023), which performs Metropolis-Hastings sampling guided by a future regressor.

Unsupervised alignment methods include TinyStyler (Horvitz et al., 2024b), which conditions on a mean-pooled style embedding for references and utilizes self-distillation, and ASTRAPOP (Liu et al., 2024), a policy optimization conditioning on the entire reference sentences. Beyond its original length-based reward, we also train ASTRAPOP using the JOINT as a reward for fair comparison.

Implementation details. We adopt the pre-trained T5-large (Raffel et al., 2020) as base architecture and utilize PEGASUS (Zhang et al., 2019a) to paraphrase the source text. For baselines requiring style embedding as guidance, we employ the STYLE embedder (Wegmann et al., 2022). Where baselines originally use the UAR embedder (Rivera-Soto et al., 2021), we instead provide a mean-pooled reference embedding. We note the UAR embedder is reserved solely for evaluation to ensure a fair comparison. TinyStyler performs reranking of candidate outputs using the JOINT score at inference time. Following the same setup, we also perform reranking, but compute style fidelity against the style coordinate predicted by the Stylo-navigator. Other implementation details are provided in the Appendix.

[2] All-the-news: this dataset was downloaded from https://huggingface.co/datasets/rjac/all-the-news-2-1-Component-one.

Method	Reddit				Blog				News			
	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT
STYLL(Qwen2.5-7B)	0.811	0.063	0.428	0.208	0.812	0.071	0.408	0.179	0.739	0.009	0.479	0.061
GPT4-turbo	0.814	0.081	0.702	0.314	0.896	0.128	0.713	0.331	0.677	0.063	0.860	0.290
GPT5.4	0.918	0.117	0.597	0.359	0.958	0.093	0.526	0.276	0.848	0.071	0.718	0.273
Llama3.1-8B-Instruct	0.756	0.135	0.587	0.390	0.875	0.173	0.539	0.407	0.819	0.107	0.493	0.281
Qwen2.5-7B-Instruct	0.733	0.139	0.593	0.403	0.823	0.169	0.545	0.393	0.763	0.132	0.577	0.340
ParaGuide$_{\lambda=200}$	0.763	0.053	0.598	0.235	0.661	0.069	0.719	0.301	0.595	0.038	0.545	0.187
ParaGuide$_{\lambda=2500}$	0.853	0.067	0.450	0.240	0.736	0.100	0.627	0.341	0.515	0.026	0.685	0.164
StyleMC	0.658	0.036	0.450	0.154	0.565	0.063	0.439	0.195	0.403	0.029	0.574	0.153
ASTRAPOP	0.578	0.027	0.728	0.171	0.997	0.255	0.014	0.060	0.840	0.060	0.170	0.139
ASTRAPOP$_{\text{JOINT}}$	0.620	0.029	0.695	0.173	0.840	0.070	0.171	0.139	0.576	0.082	0.713	0.319
TinyStyler$_{\text{REC}}$	0.897	0.144	0.352	0.323	0.791	0.134	0.603	0.379	0.605	0.053	0.582	0.223
TinyStyler$_{\text{REC,RERANK(5)}}$	0.888	0.141	0.506	0.387	0.793	0.135	0.721	0.421	0.598	0.054	0.691	0.252
TinyStyler	0.860	0.122	0.626	0.399	0.743	0.129	0.786	0.434	0.541	0.057	0.797	0.278
TinyStyler$_{\text{RERANK(5)}}$	0.859	0.122	0.730	0.436	0.756	0.128	0.844	0.452	0.561	0.057	0.843	0.294
HyperStyler$_{\text{REC}}$	0.806	0.155	0.475	0.378	0.676	0.187	0.684	0.477	0.581	0.110	0.615	0.355
HyperStyler$_{\text{REC,RERANK(5)}}$	0.800	0.152	0.705	0.460	0.692	0.189	0.854	0.537	0.587	0.101	0.802	<u>0.399</u>
HyperStyler	0.818	0.152	0.578	**0.418**	0.731	0.183	0.701	**0.489**	0.578	0.094	0.683	**0.365**
HyperStyler$_{\text{RERANK(5)}}$	0.815	0.147	0.791	<u>0.485</u>	0.736	0.183	0.864	<u>0.538</u>	0.585	0.083	0.865	0.372

Table 1: Performance comparison results on Reddit, Blog, and News datasets. For the Reddit dataset, we report the average performance across three test splits (*Diverse*, *Random*, and *Single*). For each model, the highest JOINT scores obtained from a single generation (without reranking) are **bolded**, while the overall best scores achieved with reranking are <u>underlined</u>.

4.2 Results

Overall Performance. As shown in Table 1, HyperStyler consistently outperforms all existing baselines across the three datasets. While there exists an inherent trade-off between style fidelity and semantic preservation (Horvitz et al., 2024a,b), HyperStyler achieves a better balance than TinyStyler: although TinyStyler achieves higher SIM scores, its TOWARDS scores are consistently lower across all datasets, suggesting that its higher SIM stems from weaker stylistic transformation. HyperStyler, by contrast, improves TOWARDS while maintaining competitive SIM, resulting in consistently higher JOINT scores. When reranking is applied, further improvements are observed across metrics.

Most baselines have relatively low performance on the News dataset. This is because news articles are more formal and exhibit lower stylistic variability, which makes it more difficult to capture distinctive author-specific styles (Eder et al., 2021; Wang and Riddell, 2022). Despite this challenge, HyperStyler maintains strong performance on News, outperforming even large language models[3].

Generalization Capability. We further evaluate the generalization capability of HyperStyler against TinyStyler, the best-performing baseline in our experiment. We first evaluate cross-domain transfer where either the input text or the target author belongs to an unseen domain. We then examine cases where both the input text and the target author belong to unseen domains. For **Reddit**, we use the *random* split. All results are reported without reranking.

Table 2 shows that HyperStyler achieves consistently strong performance across most domain pairs. In contrast, TinyStyler shows a consistent drop when transferring from the News domain to the more stylistically variable Blog and Reddit domains (Wang et al., 2025). This suggests that a single author embedding may be insufficient to capture context-dependent stylistic variation. Notably, HYPERSTYLER remains stable in both transfer directions under these conditions without pronounced degradation.

We also evaluate in-domain authorship style transfer under out-of-domain training conditions. As shown in Table 3, TinyStyler performs well when the transformation is confined to its training domain, but its performance degrades substantially when applied to other domains. In contrast, HyperStyler exhibits relatively limited degradation under these settings. Interestingly, a HyperStyler trained on the News dataset achieves Blog→Blog transfer performance close to that of TinyStyler trained on the Blog dataset. Overall, this result highlights the robust generalization capability of HyperStyler.

[3] GPT-5.4 underperforms other large language models, possibly because logical reasoning alone cannot fully derive an author's context-dependent style patterns from sparse references.

Model	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT
	Train: Reddit															
	Reddit → Blog				Reddit → News				Blog → Reddit				News → Reddit			
TinyStyler	0.768	0.175	0.653	0.477	0.716	0.123	0.621	0.412	0.672	0.093	0.797	0.393	0.428	0.037	0.843	0.231
HyperStyler	0.757	0.239	0.601	**0.499**	0.718	0.182	0.567	**0.446**	0.755	0.209	0.604	**0.481**	0.660	0.197	0.614	**0.459**
	Train: Blog															
	Blog → News				Blog → Reddit				News → Blog				Reddit → Blog			
TinyStyler	0.632	0.103	0.759	0.399	0.678	0.094	0.794	**0.400**	0.469	0.030	0.835	0.224	0.765	0.176	0.662	0.479
HyperStyler	0.630	0.117	0.720	**0.411**	0.618	0.104	0.718	0.397	0.596	0.154	0.718	**0.442**	0.762	0.252	0.639	**0.523**
	Train: News															
	News → Blog				News → Reddit				Blog → News				Reddit → News			
TinyStyler	0.470	0.031	0.837	0.230	0.426	0.037	0.846	0.237	0.633	0.101	0.754	0.396	0.713	0.123	0.613	0.412
HyperStyler	0.516	0.100	0.737	**0.391**	0.444	0.066	0.725	**0.315**	0.714	0.193	0.593	**0.454**	0.779	0.250	0.507	**0.468**

Table 2: Cross-domain authorship style transfer performance across three datasets. *Source* → *Target* indicates the transformation of texts from a source domain author's style to a target domain author's style. For each source-target pair, the highest JOINT scores are highlighted in bold. Values shaded in red denote cases where the JOINT score is lower than 0.3.

Model	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT	AWAY	TOWARDS	SIM	JOINT
	Train: Reddit											
	Reddit → Reddit				Blog → Blog				News → News			
TinyStyler	0.860	0.122	0.626	0.399	0.776	0.111	0.733	0.388	0.600	0.017	0.762	0.123
HyperStyler	0.818	0.152	0.578	**0.418**	0.779	0.172	0.669	**0.476**	0.601	0.045	0.724	**0.249**
	Train: Blog											
	Reddit → Reddit				Blog → Blog				News → News			
TinyStyler	0.828	0.061	0.711	0.276	0.743	0.129	0.786	0.434	0.538	0.034	0.800	0.216
HyperStyler	0.787	0.073	0.641	**0.304**	0.731	0.183	0.701	**0.489**	0.556	0.057	0.746	**0.284**
	Train: News											
	Reddit → Reddit				Blog → Blog				News → News			
TinyStyler	0.718	0.029	0.726	0.182	0.673	0.046	0.780	0.243	0.541	0.057	0.797	0.278
HyperStyler	0.766	0.036	0.614	**0.185**	0.694	0.120	0.712	**0.428**	0.578	0.094	0.683	**0.365**

Table 3: In-domain authorship style transfer performance across Reddit, Blog, and News datasets. For each setting, the highest JOINT scores are highlighted in **bold**. We shade the values representing the best-performing results within each training domain: red for TinyStyler and yellow for HyperStyler.

4.3 Ablation Study

We investigate the individual contribution of each component within HyperStyler. For this analysis, we use three test splits of the Reddit dataset. Table 4 shows our overall ablation results.

Effect of the Stylo-navigator. We compare our explicit and dynamic style selection against two alternative strategies: providing a static embedding and relying on an implicit selection. (1) **Mean pooling**: We inject the mean-pooled reference embedding into the Stylo-HyperNet. (2) **Cross-attention**: We provide all reference embeddings and perform the layer-wise style selection via cross-attention, instead of the bilinear interaction. The results demonstrate that both static embeddings and implicit selection mechanisms are less effective at capturing nuanced stylistic signals, whereas Stylo-navigator precisely selects the most relevant features.

Effect of the Stylo-hyperNet. We also compare parameter modulation against two hidden-state injection strategies:(1) **Global**: We concatenate predicted stylistic coordinate z to the encoder hidden states, providing the same style signal to all decoder layers. (2) **Layer-wise**: We generate layer-specific style embeddings using the same bilinear interaction as our method, but instead of modulating the parameters, we concatenate these embeddings to the encoder hidden states for each corresponding decoder layer. The results show that the global strategy leads to poor performance, failing to induce meaningful stylistic changes. This suggests that precise style control requires fine-grained, layer-wise adjustments rather than a single global signal. While layer-wise injection shows some improvement, it still falls short of our parameter modulation, highlighting that weight modulation is a more powerful mechanism for effective stylistic transformation.

Model	AWAY	TOWARDS	SIM	JOINT
HyperStyler	0.818	**0.152**	0.578	**0.418**
w/o Stylo-navigator (Mean-pooling)	0.783	0.099	0.706	0.368
w/o Stylo-navigator (Cross-attention)	0.778	0.114	0.671	0.384
w/o Stylo-hypernet (Global)	0.990	0.006	0.165	0.016
w/o Stylo-hypernet (Layer-wise)	0.791	0.121	0.630	0.394
w/o adapter in FFN	0.800	0.134	0.608	0.409
w/o prefixs in CrossAttn	0.825	0.150	0.551	0.408
w/ prefixs in SelfAttn	0.970	0.016	0.460	0.082
Underlying paraphraser	0.896	0.013	0.702	0.314

Table 4: Ablation study on HyperStyler. We report the averaged value across three test splits of the Reddit.

Model	r	p	AWAY	TOWARD	SIM	JOINT	#Params	Δ	t
TinyStyler	-	-	0.860	0.122	0.626	0.399	783M	-	1.28
HyperStyler	1	1	0.801	0.141	0.602	0.413	802M	+2.4%	1.34
	8	5	0.817	0.148	0.580	0.414	817M	+4.3%	1.44
	32	5	0.818	0.152	0.578	**0.418**	867M	+10.7%	1.50

Table 5: Parameter efficiency analysis. r and p denote the adapter rank and prefix length, respectively. #Params denotes the number of parameters and Δ represents the percentage increase relative to the T5-large model, and t denotes the average inference time (s) per instance.

Impact of Modulation Placement. We investigate the optimal integrations for Stylo-hyperNet by comparing different combinations of modulated components. Our results show that modulating only the cross-attention (*w/o Adapter in FFN*) achieves high performance, yet it falls short of the full model's stylistic intensity. Conversely, modulating only the FFN (*w/o Prefixs in CrossAttn*) increases stylistic scores but leads to lower content preservation. Meanwhile, we observe that modulating the self-attention layers results in broken sentence structures, as it disrupts the decoder's generation process. These findings confirm that our final configuration (modulating both cross attention and FFN) is the optimal setup.

Performance vs. Parameter Trade-off. We evaluate the parameter efficiency of our approach by testing a constrained configuration with a reduced rank and prefix length. As summarized in Table 5, decreasing the number of parameters leads to a slight degradation in performance, particularly with minor decreases in AWAY and TOWARDS. Nevertheless, even with the rank and prefix length set to 1, our model still outperforms TinyStyler while adding only 2.4% of the underlying model's parameters. This result demonstrates that HyperStyler maintains robust performance even under parameter constraints. Moreover, HyperStyler introduces minimal inference overhead compared to TinyStyler, demonstrating its practical efficiency.

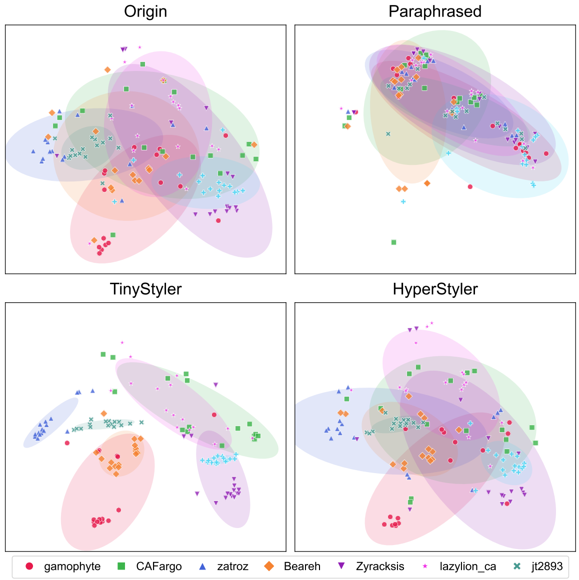

Figure 2: Visualization

Context-dependent Style Realization By paraphrasing the original texts and then reconstructing them, we analyze whether the model can recover the original style while accounting for the source context. As shown in Figure 2, TinyStyler, which conditions generation on a single static embedding, fails to faithfully reproduce the distribution of each author's original texts. In contrast, HyperStyler closely matches the original stylistic distribution, demonstrating that it can faithfully realize each author's context-aware stylistic variation.

5 Conclusion

We introduce HyperStyler, a novel architecture that explicitly decomposes LAST into two stages, context-aware style inference and stylistic manifestation, handled by specialized modules. Our extensive evaluations demonstrate that HyperStyler consistently outperforms existing baselines and exhibits robust cross-domain generalization across Reddit, Blog, and News. Notably, even in a parameter-constrained setting with only a 2.4% increase in parameters, HyperStyler maintains superior performance, showing its efficiency and effectiveness. Our design is motivated by stylometric findings that authorship is context-dependent rather than static. Our ablation studies suggest that implicit style inference and single static author embeddings are limited in capturing intra-author stylistic variability. We believe our findings point to a promising direction for future research that integrates stylometric principles into LAST, enabling more nuanced and personalized text generation.

6 Limitations

Despite the superior performance of HyperStyler across three distinct domains, we acknowledge several limitations inherent in the current LAST task setting.

Our study primarily focuses on single-sentence transformation, typically within 60–80 tokens, following the established protocols of the LAST benchmark (Patel et al., 2022). In real-world scenarios, stylistic editing often occurs at the paragraph or document level. However, expanding to longer texts is currently hindered by the lack of reliable evaluation metrics for long-form style transfer. Most existing automated metrics based on UAR provide for short-text similarity and may fail to capture cross-sentence stylistic coherence. We believe that the development of specialized evaluation frameworks for long-form LAST is a prerequisite for extending this work to broader contexts.

A second limitation is that we did not conduct evaluations with expert groups, such as forensic linguists. As noted in the foundational work on LAST (Patel et al., 2022), identifying subtle stylistic nuances is a significantly challenging task for untrained humans. Their results demonstrated that neural models (UAR, 84% accuracy) outperformed humans (77.6% accuracy) in author identification with a confidence level of $p = 0.05$. Furthermore, in evaluating style-transferred outputs, no significant difference was observed ($p = 0.05$) in identification accuracy between humans (63.4%) and neural models (62.2%). These findings suggest that the evaluation of LAST is inherently more difficult for untrained human evaluators than for specialized neural models. Therefore, while our evaluation primarily relies on automated metrics, this approach is an inevitable choice given the established difficulty of obtaining reliable and discriminative human judgments for this task.

7 Ethics Considerations

Potential Misuse and Impersonation: LAST enables effective content personalization and stylistic imitation using only a few examples. However, this technique could be exploited by malicious actors for unauthorized impersonation. High-fidelity stylistic imitation, which HyperStyler achieves, poses a significant challenge to existing AI-generated text detection methods, suggesting that a new paradigm for authorship-aware detection is required to identify sophisticated synthetic texts. We advocate for the respectful use of stylistic imitation and emphasize that the responsibility for the generated content remains with the user.

Content Risks and Potential Bias: Our training data includes datasets from online communities such as Reddit, which inherently contain offensive language, sexual content, or unethical sentiments. In this study, we did not apply explicit pre-filtering to the training data to preserve the raw stylistic features of the source domains. Consequently, the model may inadvertently generate unethical or biased outputs. We strongly advise that robust safety filters and post-processing mechanisms must accompany any real-world deployment of the model to prevent the dissemination of harmful content.

Acknowledgments

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A Data Description

We use Million User Dataset (MUD) (Khan et al., 2021)), a large-scale user text dataset collected from the social media platform Reddit. The dataset comprises over 300 million Reddit posts produced by approximately one million users over the course of one year, and includes text-based user contributions in the form of comments. Evaluation is conducted on three predefined splits from (Patel et al., 2022).

- **Diverse**: Source and target authors with posts on diverse topics across 13 or more different subreddits.
- **Random**: Random source and target authors.
- **Single**: All posts belong to a popular college football subreddit.

A.1 Blog

We use a personal blog corpus (Schler et al., 2006) collected from blogger.com. The corpus consists of blog posts written by 19,320 individual bloggers.

A.2 News

The dataset contains news articles and associated metadata collected from major U.S. and English-language news outlets, including *The New York Times*, *The Guardian*, and *Vox*.

Since our objective is to analyze writing characteristics at the single-author level, we apply a series of data filtering steps. **First**, articles with missing author information are removed. **Second**, articles whose author field refers to an organization, group, or other non-individual entity are excluded. **Third**, articles attributed to multiple authors are discarded to ensure analytical consistency.

After filtering, only articles written by clearly identifiable single human authors are retained for our experiments.

Dataset	#Samples	#Authors	#Synthetic parallel pairs
Reddit	7.5M	946k	160K
Blog	177K	17K	32K
News	538K	53K	160K

Table 6: Dataset statistics for Training

B Implementation details

All experiments are conducted using two NVIDIA A100 (80GB) GPUs. The batch size is set to 128 for all training stages, including those for the baselines.

B.1 HyperStyler

We set the adapter rank to 32 and the prefix length to 5. To derive style-conditioned embeddings, we employ a multi-head decomposition (Vaswani et al., 2017) with $H = 16$ to increase expressivity and set the dimension $d = d_e$ for simplicity. We share the projection matrices across modulation targets and embeddings for parameter efficiency.

Configuration	**Stage 1**	**Stage 2**	**Stage 3**
Learning rate	$5e^{-5}$	$1e^{-4}$	$1e^{-6}$
Scheduler	Constant	Cosine	Constant
Warm-up steps	2000	–	–
Max steps / epochs	200K steps	100K steps	3 epochs
Weight decay	0.01	0.01	0.01

Table 7: Training setup for HyperStyler

B.2 STYLEMC

Since the official source code for StyleMC (Khan et al., 2023) is not publicly available, we implemented the method by strictly adhering to the descriptions provided in the original paper. While we made every effort to ensure a faithful reproduction, minor discrepancies may exist compared to the original implementation due to unspecified details of the algorithm or hyperparameters. For a fair comparison, we modified the baseline STYLEMC by replacing its original UAR-based implementation with a STYLE Embedder (Wegmann et al., 2022). An author's representative embedding was then calculated via mean pooling, aligning it with the evaluation protocol used for other models.

Configuration	Value
Learning rate	$1e^{-5}$
Batch size	128
Optimizer	AdamW
Weight decay	0.01
Style embedding	AnnaWegmann/Style-Embedding
Underlying LM	"facebook/opt-1.3b"
Number of steps	$80 \times$ length of the sequence
α_{fluency}	0.005
α_{style}	1.0
α_{semantic}	1.0
α_{edit}	0.01

Table 8: Training settings for the Feature Regressor and sampling parameters for StyleMC.

B.3 ASTRAPOP

We conduct experiments based on the ASTRAPOP framework (Liu et al., 2024) with CPO (Xu et al., 2024), its best-performing variant, while modifying several components to better align it with our experimental setting. Originally, ASTRAPOP uses LLaMA-7B as its backbone architecture. To ensure a fair and controlled evaluation, we replace the backbone with T5-Large, which follows an encoder–decoder architecture and is consistently used across our experiments. Furthermore, replacing the backbone architecture may lead to training instability when directly applying the original reward formulation used in ASTRAPOP. To mitigate this issue, we conduct additional experiments that adopt a JOINT reward formulation instead of relying on the original reward.

To examine the impact of these architectural modifications, we evaluate four configurations with concise and controlled design variations. The experimental results are summarized in Table 12.

- **ASTRAPOP**: The baseline configuration, which follows the original input formulation and reward design.
- **ASTRAPOP$_{\textbf{JOINT}}$**: This configuration adopts the JOINT reward formulation while preserving the original input order.
- **ASTRAPOP$_{\textbf{reverse}}$**: In this setting, the `[src]` token is placed at the beginning of the input sequence, while the original reward formulation remains unchanged.
- **ASTRAPOP$_{\textbf{reverse, JOINT}}$**: This version combines both the reversed input order and the JOINT reward.

B.4 GPT-based Baselines

We use GPT-4-turbo, GPT5.4 (Medium) with the default settings provided by OpenAI.

B.5 STYLL

We followed the experimental setup used in previous studies (Patel et al., 2022) , except that we replaced the model with Qwen2.5-7B.

B.6 Other Baselines

We followed the experimental setup used in previous studies (Horvitz et al., 2024a,b).

Configuration	SFT	CPO
learning rate	$5e^{-5}$	$1e^{-5}$
batch size	128	128
# epochs	20	20
Max steps	100K	100K
β	–	0.1
top p	–	1.0
temperature	–	0.8
length penalty α	–	0.5
Optimizer	Adam	Adam
Context Size	512	512
Output Size	80	80

Table 9: Hyperparameters of ASTRAPOP.

Configuration	Value
Pretrained Ckpt	xhan77/ssdklm
Learning rate	5×10^{-6}
Batch size	128
Optimizer	AdamW
Weight decay	0.01
Schedule	Constant
Diffusion Steps	200
Context Size	80
Output Size	80
Warm-up Steps	2000
Total Steps	150K

Table 10: Training PARAGUIDE hyperparameters for Blog, News, and Reddit dataset.

Configuration	Value
Pretrained Ckpt	google/t5-v1_1-large
Learning rate	$1e^{-5}$
Batch size	128
Optimizer	Adam
Weight decay	0.01
Schedule	Constant
Warm-up Steps	2000
Total Steps	150K

Table 11: Training Tinystyler hyperparameters for Blog, News, and Reddit dataset.

C Evaluation Details

C.1 Metric Formula Definition

We denote the set of source authors as S and the set of target authors as T. For any author a, we represent the set of their 16 posts as P_a, and we denote the set of 16 posts written by a source author s and style-transferred to the target author t as $P_{s \to t}$. We let $\mathbf{R}(P)$ denote a single UAR embedding produced over a set of posts P. The average Mutual Implication Score between two sets of posts authored by a and b is denoted as $\mathrm{MIS}(P_a, P_b)$. Finally, we define $S(\mathbf{u}, \mathbf{v})$ scaled to the range $[0, 1]$, given by

$$S(\vec{u}, \vec{v}) = \frac{\mathrm{sim}(\vec{u}, \vec{v}) + 1}{2}.$$

We further define its complement as

$$S_c(\vec{u}, \vec{v}) = 1 - S(\vec{u}, \vec{v}).$$

Away :Measures how far a style-transferred text departs from the source author's style.

$$\mathrm{AWAY}(s,t) = \frac{\min\Big(S_c\big(\vec{R}(P_{s\to t}), \vec{R}(P_s)\big),\ S_c\big(\vec{R}(P_t), \vec{R}(P_s)\big)\Big)}{S_c\big(\vec{R}(P_t), \vec{R}(P_s)\big)} \tag{10}$$

Towards : Measures how far a style-transferred text moves toward the target author's style.

$$\mathrm{TOWARDS}(s,t) = \frac{\max\Big(S\big(\vec{R}(P_{s\to t}), \vec{R}(P_t)\big) - S\big(\vec{R}(P_s), \vec{R}(P_t)\big), 0\Big)}{S_c\big(\vec{R}(P_s), \vec{R}(P_t)\big)} \tag{11}$$

Sim :Measures how well the source text preserves its meaning while being transformed into the target author's style.

$$\mathrm{SIM}(s,t) = \frac{\max\Big(\mathrm{MIS}(P_{s\to t}, P_s) - \mathrm{MIS}(P_t, P_s), 0\Big)}{1 - \mathrm{MIS}(P_t, P_s)} \tag{12}$$

Joint : Summarizes overall performance by aggregating multiple evaluation metrics using their geometric mean.

$$\mathrm{JOINT}(s,t) = G\Big(G\big(\mathrm{AWAY}(s,t), \mathrm{TOWARDS}(s,t)\big), \mathrm{SIM}(s,t)\Big) \tag{13}$$

Method	Reddit				Blog				News			
	Away	Towards	Sim	Joint	Away	Towards	Sim	Joint	Away	Towards	Sim	Joint
ASTRAPOP	0.578	0.026	0.728	0.166	0.997	0.255	0.014	0.060	0.840	0.060	0.170	0.139
ASTRAPOP$_{\text{JOINT}}$	0.619	0.028	0.714	0.172	0.840	0.070	0.171	0.139	0.576	0.082	0.713	0.319
ASTRAPOP$_{\text{REVESRSE}}$	0.571	0.026	0.743	0.172	0.991	0.275	0.077	0.170	0.655	0.086	0.394	0.248
ASTRAPOP$_{\text{REVESRSE_JOINT}}$	0.596	0.026	0.725	0.172	0.988	0.243	0.125	0.213	0.950	0.005	0.202	0.115

Table 12: Performance comparison results on the configuration of ASTRAPOP.

Method	Diverse				Random				Single			
	Away	Toward	Sim	Joint	Away	Toward	Sim	Joint	Away	Toward	Sim	Joint
STYLL(Qwen2.5-7B)	0.797	0.069	0.405	0.200	0.739	0.074	0.416	0.240	0.896	0.047	0.463	0.184
gpt-4.1-2025-04-14	0.854	0.086	0.712	0.318	0.797	0.088	0.754	0.359	0.894	0.084	0.708	0.329
gpt-5.4-2026-03-05 (medium)	0.94	0.077	0.501	0.238	0.866	0.139	0.679	0.436	0.948	0.135	0.61	0.402
meta-llama/Llama-3.1-8B-Instruct	0.724	0.119	0.602	0.371	0.743	0.127	0.564	0.369	0.8	0.158	0.595	0.431
Qwen/Qwen2.5-7B-Instruct	0.699	0.115	0.615	0.377	0.675	0.147	0.632	0.427	0.824	0.154	0.533	0.405
ParaGuide$_{\lambda=200}$	0.774	0.056	0.544	0.221	0.696	0.047	0.585	0.222	0.818	0.057	0.664	0.263
ParaGuide$_{\lambda=2500}$	0.859	0.078	0.381	0.239	0.801	0.065	0.456	0.246	0.900	0.058	0.512	0.235
StyleMC	0.603	0.051	0.462	0.189	0.625	0.039	0.453	0.173	0.746	0.017	0.435	0.100
ASTRAPOP	0.612	0.031	0.679	0.168	0.516	0.027	0.736	0.192	0.607	0.022	0.770	0.152
ASTRAPOP$_{\text{JOINT}}$	0.656	0.035	0.645	0.168	0.550	0.029	0.702	0.205	0.653	0.022	0.739	0.145
TinyStyler$_{\text{REC}}$	0.897	0.130	0.306	0.282	0.863	0.154	0.314	0.315	0.932	0.148	0.436	0.371
TinyStyler$_{\text{REC,RERANK(5)}}$	0.883	0.127	0.462	0.346	0.854	0.148	0.465	0.384	0.927	0.149	0.592	0.432
TinyStyler	0.836	0.109	0.582	0.356	0.835	0.127	0.590	0.396	0.910	0.130	0.706	0.445
TinyStyler$_{\text{RERANK(5)}}$	0.831	0.111	0.693	0.393	0.839	0.125	0.705	0.434	0.907	0.131	0.793	0.480
HyperStyler$_{\text{REC}}$	0.810	0.162	0.426	0.368	0.753	0.163	0.460	0.377	0.854	0.139	0.540	0.390
HyperStyler$_{\text{REC,RERANK(5)}}$	0.802	0.157	0.649	0.444	0.749	0.161	0.706	0.469	0.849	0.138	0.760	0.468
HyperStyler	0.813	0.153	0.547	0.402	0.768	0.157	0.542	0.410	0.872	0.145	0.645	0.443
HyperStyler$_{\text{RERANK(5)}}$	0.807	0.149	0.760	**0.467**	0.766	0.148	0.768	**0.480**	0.871	0.145	0.844	**0.508**

Table 13: Reddit Diverse/Random/Single split results

Model	Diverse				Random				Single			
	Away	Toward	Sim	Joint	Away	Toward	Sim	Joint	Away	Toward	Sim	Joint
HyperStyler	0.813	**0.153**	0.547	**0.402**	0.768	**0.157**	0.542	**0.410**	0.872	**0.145**	0.645	**0.443**
w/o Stylo-navigator (mean-pooling)	0.776	0.088	0.677	0.332	0.732	0.112	0.666	0.389	0.842	0.097	0.775	0.385
w/o Stylo-navigator (Cross-attention)	0.770	0.102	0.641	0.348	0.729	0.124	0.627	0.392	0.833	0.116	0.746	0.412
w/o Stylo-hypernet (Global)	0.985	0.009	0.142	0.015	0.986	0.008	0.150	0.033	0.999	0.000	0.202	0.002
w/o Stylo-hypernet (Layer-wise concat)	0.789	0.119	0.597	0.374	0.734	0.123	0.595	0.388	0.852	0.121	0.697	0.421
w/o adapter in FFN	0.794	0.129	0.579	0.385	0.747	0.139	0.565	0.400	0.859	0.134	0.681	0.443
w/o prefixs in CrossAttn	0.830	0.150	0.521	0.396	0.768	0.156	0.512	0.399	0.877	0.142	0.621	0.430
w/ prefixs in SelfAttn	0.958	0.022	0.416	0.094	0.960	0.016	0.434	0.081	0.990	0.010	0.529	0.071
Underlying paraphraser	0.889	0.019	0.682	0.123	0.849	0.012	0.696	0.084	0.949	0.007	0.776	0.058

Table 14: Ablation study results on Reddit (Diverse / Random / Single) splits

HyperStyler HyperStyler	HyperStyler HyperStyler	Diverse				Random				Single			
Rank	Prefix	Away	Toward	Sim	cJoint	Away	Toward	Sim	Joint	Away	Toward	Sim	cJoint
1	1	0.804	0.140	0.577	0.398	0.745	0.141	0.552	0.396	0.854	0.142	0.676	**0.445**
8	5	0.819	0.147	0.551	0.400	0.763	0.154	0.540	0.403	0.870	0.143	0.648	0.440
32	5	0.813	0.153	0.547	0.402	0.768	0.157	0.542	0.410	0.872	0.145	0.645	0.443
128	5	0.822	**0.156**	0.524	**0.403**	0.770	**0.164**	0.528	**0.411**	0.877	**0.148**	0.627	0.440

Table 15: Parameter efficiency analysis on Reddit (Diverse / Random / Single) splits

Table 16: Qualitative example of style transfer (1/3). Target texts are ranked by style embedding similarity (S_{sty}) to the HyperStyler output.

Source Text	30-74 items are objects that will cause some suspicion, library cards, reading glasses, books that aren't anything to do with sports and the like.
HyperStyler Output (Joint: 0.684)	omg, the 3074 items are objects that will cause some suspicion.. library cards, reading glasses, books that have nothing to do with sports n the like
Target Texts (ranked by S_{sty})	**#1 (.977)** ng mga finalists.. ung naka-white and yellow cla.. ang galing tlaga.. how i wish, makagawa rin ako ng ganun,,, JOKE!!! **#2 (.965)** marami akong tanong na hindi ko masagot, na hindi rin masagot ng mga kaibigan ko!! **#3 (.959)** i dont want anyone to know my weakness (well..except for the chosen few!! **#4 (.955)** he approached me coz i was alone while waiting for my friend, jodette!! **#5 (.947)** i didnt take it badly though.. because he reasoned out that there was traffic... **#6 (.937)** never limit urself.. spread your wings!! **#7 (.936)** i was sarcastic.. coz i was reviewing.. and it was just wrong timing.. to my mom: im really sorry... **#8 (.923)** midterms i'm gonna start reviewing for my midterms.. i cant afford to fail a test... **#9 (.923)** ahhh.. naaalala ko na!! **#10 (.919)** if there's one way i would like to express my thoughts and ideas.. it is through writing!! **#11 (.890)** she's a very responsible editor.. hail to krishna!! **#12 (.890)** natapos ko na rin ung book na pinahiram ni camille.. maganda ung story... how i wish may michael moscovitz din ako!! **#13 (.889)** and that's a record .. i had a temporary vacation from blogging because of my preliminary exams!! **#14 (.882)** we're not gonna have pe next week.. we're not gonna have pe next week!!! **#15 (.709)** 3. my skin gets itchy!! **#16 (.068)** 'Once in a while, right in the middle of an ordinary life, LOVE gives us a fairy tale!!'
Style features: double-dot pauses (..) · repeated exclamation (!! !!!) · informal abbreviations (dont, urself)	
Baseline Outputs (Joint Score)	
Qwen (.648)	so these 30-74 items are like things that will cause suspicion.. library cards.. reading glasses.. and books that have nothing to do with sports!!
LLaMA (.641)	i mean, items 30-74 are things that'd raise eyebrows.. like library cards, reading glasses, books that have nothin to do with sports!!
GPT-5.4 (.614)	well.. these 30-74 items are stuff that will make ppl suspicious.. library cards, reading glasses, books na walang kinalaman sa sports!!
TinyStyler (.607)	Items from 30 to 74 are objects that will make people suspicious.
ParaGuide (.557)	Psee-Edes... Pasia.
Styll-Qwen (.398)	Items 30-74 are objects that will cause suspicion, including library cards and reading glasses.

Table 17: Qualitative example of style transfer (2/3). HyperStyler's output most closely matches the diary-like, stream-of-consciousness reference texts.

Source Text	What do you have to do to that even though things may go wrong, you can still look back and still be proud of yourself.
HyperStyler Output **(Joint: 0.609)**	it may go wrong, but you can still be proud of yourself, even though things may go wrong...oh well, guess what you have to do...ah, yeah, u have to do
Target Texts (ranked by S_{sty})	**#1 (.952)** so now i am hoping to make posting a regular habit on here...we'll see how that goes. **#2 (.930)** been mostly OP today, minus a dorito attack, extra rice at dinner, and some wine right now...oh well **#3 (.918)** so by the time i got home around lunch, dad had been on the phone with comcast... **#4 (.895)** if it weren't for niki (again) acting like a total idiot, it would have been perfect... **#5 (.889)** so I read about this online, and decided to give it a try...dealing mostly with food stuff. **#6 (.867)** so this also means that the old ass comp got put upstairs in 'dad's' room... **#7 (.827)** i even managed to finally sift through and delete all my extraneous music on the comp **#8 (.827)** thank god the comp was fixed today...i went into ITP to watch mom do two evals... **#9 (.820)** makes me happy to find good shit like that! **#10 (.818)** got my new version of winamp, and it's smooth sailing! **#11 (.794)** 'let not fear make strangers of those who should be lovers'— easier said than done **#12 (.731)** (this is WAY not normal for me! **#13 (.699)** lol) we're waiting for stephanie to come back so we know what the heck we should be doing... **#14 (.673)** damn you ashley and billy wherever you are! **#15 (.623)** it is quite exciting i must say... so i am getting back OP this week **#16 (.228)** So let's see if this posts, and here goes!
Style features: consistent all-lowercase · trailing ellipsis (...) · casual exclamation	
Baseline Outputs (Joint Score)	
TinyStyler (.603)	even though things go wrong, you can still look back and still be proud of yourself...
LLaMA (.580)	even when things go sideways, you gotta be able to look back and still feel good about what you did... that's the real challenge.
Qwen (.516)	so ya know, even though things might go south, you can still look back and feel proud of yourself... i guess that's something, right?
ParaGuide (.491)	i'm looking for what i'd had...
GPT-5.4 (.481)	so basically...what do you gotta do so that even if things go wrong, you can still look back and be proud of yourself?
Styll-Qwen (.369)	what do you need to do so that even when things go wrong, you can look back and be proud of yourself?

Table 18: Qualitative example of style transfer (3/3). HyperStyler's "(sigh...)" interjection closely matches the target author's parenthetical asides and direct reader address.

Source Text	only 1 more month of summer this is really depressing!
HyperStyler Output (Joint: 0.651)	(sigh...) Only 1 more month of summer which sucks.
Target Texts (ranked by S_{sty})	**#1 (.883)** Until My next blog.....keep Reading this one. **#2 (.745)** *ahem* I am no more interesting then anybody reading this. **#3 (.662)** ...That I'm more interesting than most people, Probably including you. **#4 (.496)** It's close who cares, Let's move on. **#5 (.483)** After much trial and error the 3 found a permenant drummer in Andy Hurley. **#6 (.436)** Thats it, I'm out for now. **#7 (.425)** In Fact in Interestingness levels you're kicking my proverbial ass. Thank you, Now you may continue to my new Blog entry. **#8 (.397)** I'll just wrap this up with PLease, for the love of god, Everybody take a few minutes of your day and Listen to Drowning Lessons... **#9 (.385)** I don't have much to say right now cause it sucked and I don't feel like boring you as of right now. **#10 (.370)** Very captivating voice. **#11 (.358)** Of A new school year. **#12 (.357)** Now carry on with your worthless days. **#13 (.311)** Music is the best thing going, as long as you listen to the right kind of music. **#14 (.269)** Listen to something where it is possible for the artists to actually be artists and create something that took talent... **#15 (.222)** After a while, they stumbled across Patrick Stump who has a very charismatic voice, and thus, the birth of Fall Out Boy. **#16 (.155)** Todays topic of discussion (by me) on Lets see...Music!!!
Style features: extended ellipsis / parenthetical aside · direct reader address · interjections (*ahem*)	
Baseline Outputs (Joint Score)	
TinyStyler (.495)	Only 1 more month of summer, that is really depressing.
GPT-5.4 (.449)	ugh only 1 more month of summer left... this is SO depressing.
Qwen (.435)	only 1 more month of summer... *ahem* this is really depressing!
Styll-Qwen (.430)	Only 1 more month of summer! This is really depressing!
ParaGuide (.419)	Hooray only 1 more month of summer...
LLaMA (.392)	*sigh* only 1 more month of summer left, this is pretty depressing ngl.